

Chapter - 15

Environmental Factors and Crop Physiology: Influence of Light, Temperature and CO₂ on Crop Performance

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Abstract

Environmental factors, including light, temperature, and carbon dioxide (CO₂) levels, play a crucial role in shaping crop physiology and performance. These factors have complex and interdependent interactions that significantly influence crop growth, development, and productivity. Light is essential for plant growth as it drives photosynthesis, providing the energy required for various metabolic processes and regulating plant growth and development. Temperature influences various stages of plant growth, affecting germination, photosynthesis, respiration, and reproductive success. Elevated CO₂ levels stimulate photosynthesis and enhance plant growth, with differential responses observed between C3 and C4 plants. Additionally, elevated CO₂ levels can impact plant water relations, leading to reduced stomatal conductance and transpiration. However, under heat stress, elevated CO₂ can partially close stomata, mitigating water loss. Understanding the combined effects of elevated CO₂ and temperature on crop performance is vital in the context of climate change. Optimizing environmental factors can help maximize crop productivity and minimize

stressors for sustainable agriculture.

Keywords: CO₂, light, temperature, Climate change, photosynthesis & respiration

Introduction

Environmental factors play a crucial role in crop physiology and can significantly influence crop performance. Among the key environmental factors, light, temperature, and carbon dioxide (CO₂) levels have a profound impact on crop growth, development, and productivity. It is worth noting that the interaction between these factors is complex, and their influence on crop performance is interdependent. To optimize crop performance, it is crucial to understand the specific requirements of different crops and manage the environmental factors accordingly. This knowledge can help farmers and researchers develop strategies to optimize light exposure, temperature control, and CO₂ levels to maximize crop productivity while minimizing environmental stressors. Let's explore the influence of these factors on crop performance.

Influence of Light on Crop Performance

Why light is important for plant growth?

Light (solar radiation) is crucial for plant growth because it is the primary source of energy for photosynthesis, the process by which plants convert light energy into chemical energy. Here are some reasons why light is important for plant growth:

1. **Photosynthesis:** Light energy is absorbed by pigments, primarily chlorophyll, located in plant cells. This energy is used to drive the chemical reactions of photosynthesis, where carbon dioxide and water are converted into glucose (a sugar) and oxygen. Glucose serves as a building block for plant growth and is utilized for energy production, biosynthesis of structural components, and storage.
2. **Energy production:** Through photosynthesis, light energy is converted into chemical energy in the form of adenosine triphosphate (ATP). ATP is the energy currency of cells and is required for various metabolic processes, including nutrient uptake, cellular respiration, and synthesis of molecules essential for growth and development.
3. **Carbohydrate production:** Photosynthesis also results in the production of carbohydrates, such as glucose and sucrose. Carbohydrates serve as a source of energy and carbon for plants. They are transported throughout the plant to support growth, root development, flower formation, and fruit production.
4. **Regulation of growth and development:** Light plays a critical role in

regulating plant growth and development. It affects processes such as seed germination, stem elongation, leaf expansion, root development, flowering, and fruit ripening. Different wavelengths and intensities of light can trigger specific responses in plants, influencing their morphology and physiological processes.

5. **Phototropism and photoperiodism:** Plants exhibit phototropism, which is the growth response towards or away from light. This enables plants to orient their growth towards optimal light conditions. Additionally, light duration or photoperiod influences crucial plant processes, such as flowering, dormancy, and the timing of developmental stages.

Plants exhibit various responses to the **quality (wavelength), intensity, and duration of radiation** they receive. These responses are collectively known as **photomorphogenic responses**, as they affect the plant's growth, development, and physiological processes. Here are some key aspects of plant responses to these factors:

Quality of Light

The quality of light refers to its spectral composition, which describes the distribution of wavelengths within the light spectrum. Figure 1 is classic description of the optical properties of a single leaf exposed to light in the wavelength's characteristics of the solar spectrum. In the visible spectrum (0.4 to 0.7 μm) approximately 85-90% light is absorbed, 5-10% is reflected, and 5% is transmitted by the canopies. Plants have a high absorption rate for blue light (approximately 0.45-0.50 μm) and red light (approximately 0.65-0.70 μm), while they reflect green light. Near infrared (NIR) (0.70 to 1.30 μm) reflection and transmission are much greater and low absorption than in visible range. If water is present on the surface of a leaf, wavelengths ranging from 1.3 μm to 2.9 μm tend to exhibit higher levels of absorption and lower levels of reflection and transmission. Based on the information provided, it can be inferred that plants primarily rely on visible light especially **blue and red light** for their internal processes and is known as photosynthetically active radiation (PAR) spectrum (**Figure 2**).

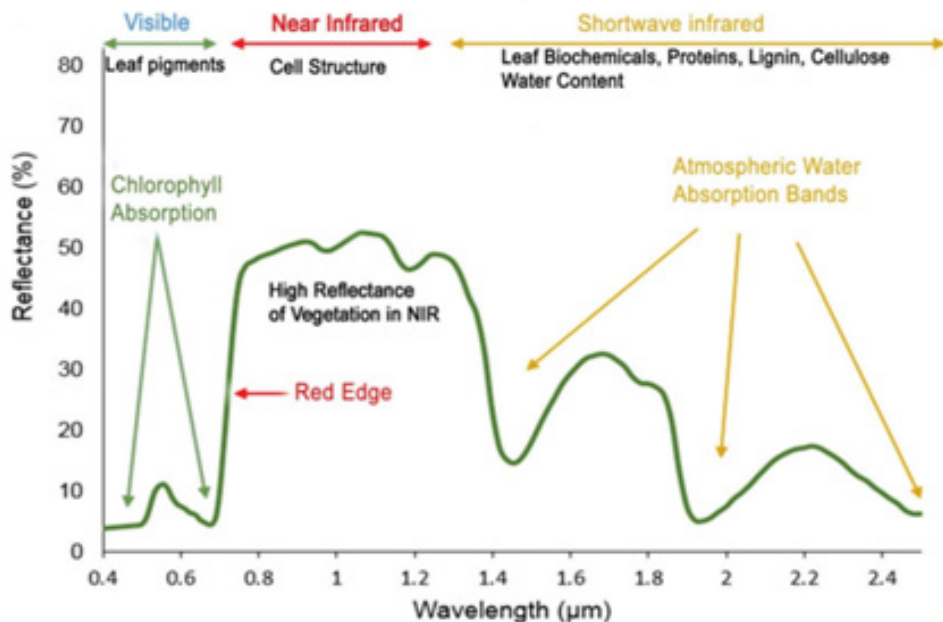


Figure 1: Spectral response of healthy & green vegetation toward optical wavelengths of light (Source: Mustafa et al., 2015)

» **Role of Blue Light (0.45 to 0.50 μm) on Plants Growth and Development**

- » **Phototropism:** Phototropism refers to the growth or movement of plant parts in response to light direction. Blue light plays a crucial role in phototropism, particularly in the bending of stems or leaves towards the light source. It influences the orientation and positioning of plant organs to capture light energy efficiently, reduce photodamage, and acquire CO₂ (Takemiya *et al.*, 2005).
- » **Stomatal regulation:** Stomata are tiny openings on the plant surface that regulate gas exchange and water vapor loss. Blue light plays a role in stomatal opening and closing, affecting the plant's water balance and gas exchange.
- » **Chloroplast Development:** Blue light is essential for the proper development and functioning of chloroplasts, the cellular structures responsible for photosynthesis. It influences chlorophyll synthesis and the formation of thylakoid membranes within chloroplasts.
- » **Flowering and Reproduction:** Blue light is involved in the regulation of flowering time and the transition from vegetative growth to reproductive development. It influences the expression of genes associated with floral initiation and flowering induction.

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- » **Photomorphogenesis:** Blue light regulates photomorphogenesis, which includes processes such as seedling development, leaf expansion, and stem elongation. It helps control plant architecture by promoting compact and sturdy growth.
 - » **Role of Red Light (0.65 to 0.70 μm) on Plants Growth and Development**
 - » **Photosynthesis:** Red light is absorbed by chlorophyll pigments in plant cells, particularly chlorophyll a, which is crucial for photosynthesis. It provides the energy needed to convert carbon dioxide and water into glucose and oxygen. Therefore, red light plays a central role in the production of carbohydrates, which are essential for plant growth.
 - » **Stem elongation:** Red light promotes stem elongation in plants, a process known as phototropism. When plants receive more red light, they tend to grow taller and develop longer internodes. This response allows plants to optimize light capture and compete for light in dense vegetation.
 - » **Flowering and fruiting:** Red light is crucial for regulating the flowering and fruiting processes in many plants. It acts as a signal for photoperiodism, the response of plants to the duration of light and dark periods. When plants receive the appropriate amount of red light during specific periods, it triggers flowering and fruiting, ensuring reproductive success.
 - » **Seed germination:** Red light also influences seed germination in some plant species. Seeds often require exposure to red light to break dormancy and initiate the germination process. Red light acts as a signal to indicate favourable conditions for seedling growth.
 - » **Chlorophyll production:** Red light promotes the synthesis and accumulation of chlorophyll in plant cells. Chlorophyll is responsible for the green colour in plants and is essential for capturing light energy during photosynthesis. Adequate levels of red-light help ensure healthy chlorophyll production, which is crucial for overall plant growth and vitality.

Wavelengths of light	Photoreceptors	Plant responses
 Red, Far-red (600-750 nm)	PHYs	Germination, De-etiolation, Shade avoidance, Inhibition of stem and petiole elongation, Leaf expansion and flattening, Circadian rhythms, Flowering, Branching.
Green (530- 570 nm)		
Blue (390-500 nm)	CRYs	De-etiolation, Inhibition of stem and petiole elongation, Leaf expansion, Circadian rhythms, Flowering, Flavonoid biosynthesis. Response to shade by B/G photoperception.
UV-A (320-390 nm)	PHOTs	Leaf flattening, Phototropism, Stomatal opening, Chloroplast relocation.
UV-B (290-315 nm)	ZTL/FKF1/LKP2	Circadian rhythms, Flowering.
	UVR8	De-etiolation, Flavonoid biosynthesis.

Figure 2: Spectral wavelength specificity of the main plant photoreceptors dictates specific plant photomorphogenesis responses (Source: Paradiso and Proietti, 2022).

Intensity of Light

The intensity of light refers to the amount of light energy present per unit area per unit time. The unit of measurement for determining the intensity of natural light is footcandles (FC). One footcandle is roughly equivalent to the brightness of one candle, located one foot away. Outdoor, direct sunlight typically has a peak intensity of around **10,000 footcandles**. The intensity of light plays a crucial role in various biological processes, including plant growth, photosynthesis, and the regulation of physiological responses.

In the context of plants, light intensity is an important factor that influences their growth and development. Here are some key aspects related to the intensity of light:

- » **Photosynthesis:** Light intensity directly affects the rate of photosynthesis in plants. As the intensity of light increases, the rate of photosynthesis generally increases until reaching a **light saturation point (Figure 3)**. Beyond this point, further increases in light intensity may not significantly enhance photosynthesis. Different plant species have varying light intensity requirements for optimal photosynthetic activity.
- » **At light compensation point;** light intensity at which the rate of photosynthesis exactly matches the rate of respiration in a plant. Below the light compensation point, the rate of respiration exceeds the rate of photosynthesis, leading to a net loss of carbon dioxide. As light intensity

increases above the compensation point, the rate of photosynthesis surpasses respiration, resulting in a net gain of carbon dioxide (**Figure 3**).

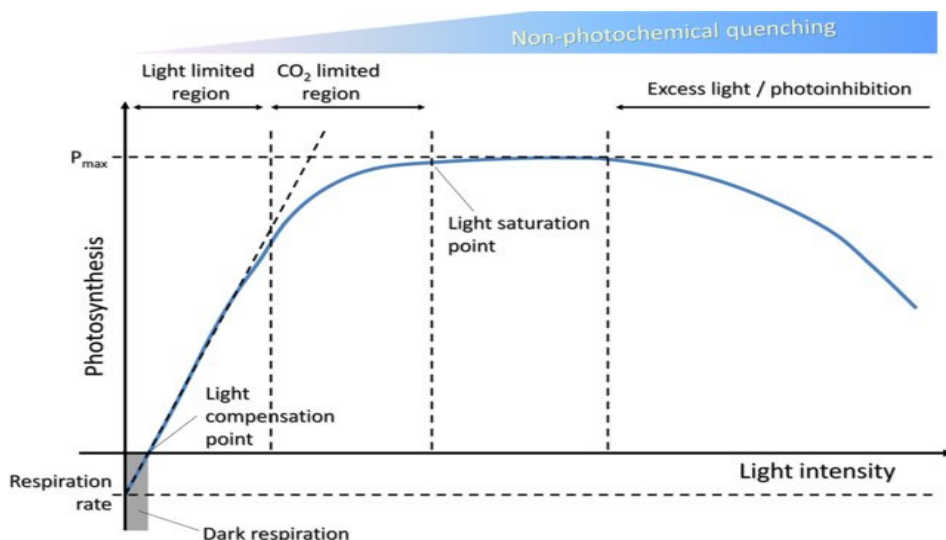


Figure 3: Light response curves for photosynthesis (Source: Benedetti et al., 2018).

- » **Plant growth and morphology:** Light intensity affects plant growth and morphology. Low light intensity can result in elongated, spindly growth and reduced leaf area. High light intensity, on the other hand, can lead to stunted growth or damage due to excessive heat or radiation. Different plant species have different tolerance ranges for light intensity, and it is important to provide an appropriate intensity level for each specific species.
- » **Flowering and fruiting:** Light intensity influences the timing and quality of flowering and fruiting in many plants. Certain plants require specific light intensity thresholds or ratios to trigger or inhibit flowering. Insufficient light intensity can result in delayed or reduced flowering, while excessively high intensity may negatively impact flower and fruit development.
- » **Light adaptation and stress responses:** Plants exhibit various adaptive responses to changes in light intensity. They can adjust their photosynthetic machinery, pigment composition, and stomatal behaviour to optimize light absorption and minimize potential damage. In response to high light intensity, plants may produce protective pigments, such as anthocyanins, or modify their leaf orientation to reduce light exposure.

Duration of Light

The duration of light refers to the length of time that plants are exposed to light within a 24-hour period, commonly known as the photoperiod. The duration of light plays a crucial role in regulating various physiological and developmental processes in plants. Here are some key aspects related to the duration of light:

- » **Photoperiodic responses:** Many plant species exhibit photoperiodic responses, which are specific physiological reactions triggered by the length of the light and dark periods. These responses can include flowering, bud formation, dormancy, and other developmental changes. **Plants can be categorized as short-day plants, long-day plants, or day-neutral plants based on their specific requirements for light duration.**
- » **Flowering induction:** The duration of light is particularly important for regulating flowering in many plant species. Short-day plants, also known as long-night plants, require a longer period of darkness than light to initiate flowering. In contrast, long-day plants, also known as short-night plants, require a shorter period of darkness than light to induce flowering. Day-neutral plants are not significantly influenced by the duration of light.
- » **Vegetative growth and dormancy:** The duration of light also influences vegetative growth and dormancy in plants. Longer light durations can promote vegetative growth and inhibit the onset of dormancy, while shorter light durations can trigger dormancy and growth cessation. The photoperiodic cues provide plants with information about seasonal changes and help regulate their growth patterns accordingly.
- » **Circadian rhythms:** Light duration is an important factor in regulating the internal biological clocks of plants, known as circadian rhythms. These rhythms govern various physiological and metabolic processes, including stomatal opening and closing, photosynthesis, hormone synthesis, and gene expression. Consistent and appropriate light durations are necessary to maintain proper circadian rhythm synchronization.

Influence of Temperature on Crop Performance

Temperature is a critical environmental factor that significantly influences crop performance and growth. **Here are some key aspects of how temperature affects crop performance:**

1. Germination and seedling establishment

Temperature plays a vital role in seed germination and subsequent seedling

establishment. Some seeds require cooler temperatures, while others need warmer conditions. The optimum temperature results in germination in the shortest time possible (Khacim *et al.*, 2022). According to certain studies, the number of seeds that germinate grows linearly as the soil temperature rises to an optimum level and then linearly declines as the soil temperature rises above the limit (Bradford, 2002), that is depicted in **Figure 4**. In general, most of plants have a preferred temperature range for germination between **18°C to 29°C**.

- » **Temperature can influence the percentage and rate of germination through three distinct physiological processes that are described below (Roberts, 1988).**
 - i. Seeds deteriorate over time due to temperature and moisture. Q10 (rate of viability loss) increases from 2 at -10°C to 10 at 70°C (The Q10 is a measure used to describe how temperature affects the rate of a biological process. In this case, it represents the rate of loss of viability in orthodox seeds). The Q10 value increases with higher temperatures, indicating that as the temperature rises, the rate of loss of viability in orthodox seeds accelerates. At -10°C, the rate is relatively slow ($Q_{10} \approx 2$), but at 70°C, the rate becomes much faster ($Q_{10} \approx 10$).
 - ii. Most seeds start in a dormant state. Dry seeds lose dormancy with increasing temperature ($Q_{10} \approx 2.5$ -3.8 up to 55°C). Hydrated seeds respond differently, with high temperatures reinforcing dormancy. Low temperatures can induce dormancy or stimulate germination (within -1°C to 15°C).
 - iii. Once seeds have lost dormancy their rate of germination shows a positive linear relation between the base temperature (at and below which the rate is zero) and the optimum temperature (at which the rate is maximal); and a negative linear relation between the optimal temperature and the ceiling temperature (at and above which the rate is again zero). The optimum germination temperature, which maximizes the germination rate, is typically higher than the temperature required to achieve the maximum percentage of germination in partially dormant or partially deteriorated seed populations.

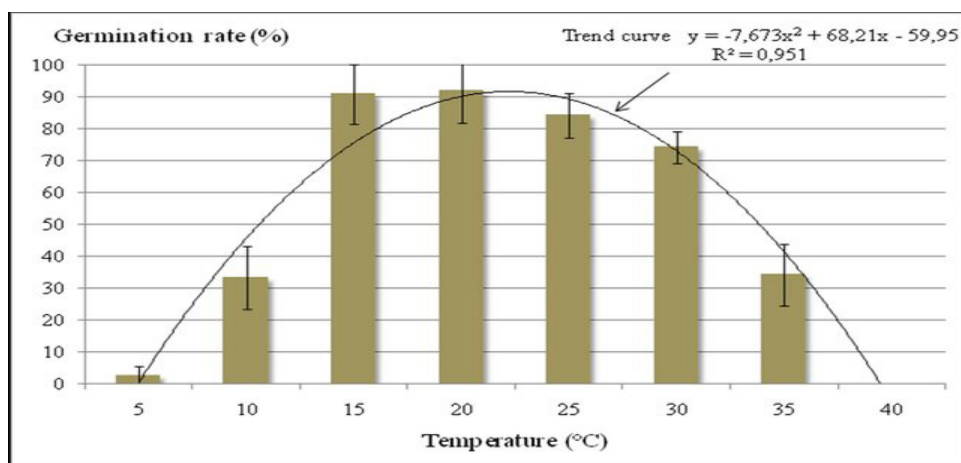


Figure 4: Effect of temperature on the germination rate of marram grass seeds (Source: Chergui et al., 2013).

1. Growth and development

Temperature affects the rate of plant growth and development. Each crop has specific maximum and minimum temperature ranges that define the limits within which observable growth occurs (Table 1). As temperatures increase towards the crops optimum level, vegetative development including the rate of node and leaf appearance, also increases. **Generally, most plant species have a higher optimum temperature for vegetative development compared to reproductive development.** Crops have specific heat unit (Growing Degree Days) requirements to complete their life cycle or progress to the next stages of growth. High temperature condition generally accelerates crops growth and shorten the time to maturity because its heat units' requirements satisfy quickly and accelerated speed of physiological, biochemical, and developmental processes in plants (**Figure 5 (a, b)**), while cooler temperatures can slow down crops growth and extend the crop cycle.

When non-perennial crops develop at a faster rate, their life cycle becomes shorter, leading to smaller plants, a reduced reproductive period, and lower yield potential. Additionally, photoperiod sensitive crops like soybeans may experience disturbances in their growth stages when exposed to temperature variations, affecting their phenological development (Hatfield and Prueger, 2015).

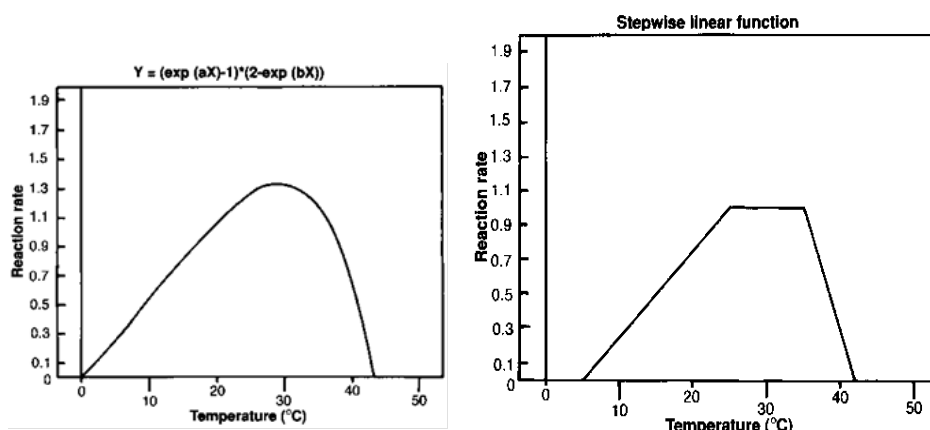


Figure 5: (a) Exponential rate of reaction as a function of temperature (b) Stepwise increase in reaction rate as a function of temperature (Source: Abrol and Ingram, 1996).

Table 1: Requirement of cardinal temperature (°C) for certain crops (Rai et al., 2013)

Crops	Minimum (base) temp.	Optimum Temp.	Maximum Temp.
Rice	10-12	30-32	36-38
Wheat	3-4.5	25	30
Maize	8-10	32-35	40-44
Bajara	8-10	30-32	40
Sorghum	8-10	32-35	40
Oat	4-5	25	30

2. Photosynthesis and carbon assimilation

Temperature influences photosynthesis, the process by which plants convert light energy into chemical energy. Within a certain temperature range, the rate of photosynthesis typically increases as temperature rises, up to an optimal point. Beyond this point, photosynthesis declines (**Figure 6 (a, b)**) due to enzyme denaturation (Rubisco), stomata closure (reduce stomatal conductance) and damage to photosystems. Extreme temperatures, both high and low, can disrupt photosynthetic processes, leading to reduced carbon assimilation and decreased crop productivity.

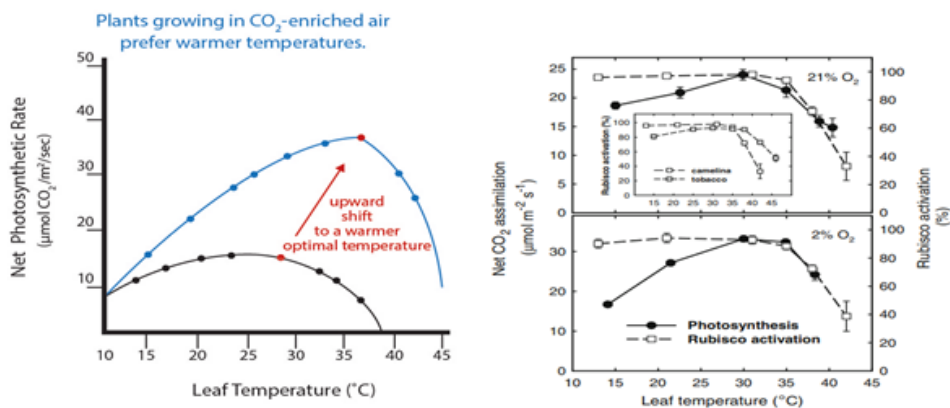


Figure 6: (a) Relationship between leaf temperature and net photosynthetic rate at ambient in *Populus grandidentata* (Source: Jurik et al., 1984), (b) Temperature response of net CO_2 assimilation rate (filled circle) and Rubisco activation (open squares) in camelina plants under photorespiratory and non-photorespiratory conditions (Source: Carmo-Silva and Saluki, 2012).

3. Respiration and energy balance

Temperature affects the rate of plant respiration; Respiration is a biochemical process that involves the breakdown of organic molecules (such as glucose) to **release energy for cellular activities**. The enzymes responsible for these reactions are sensitive to temperature changes. As the temperature rises, the activity of these enzymes also increases, leading to a higher rate of respiration and increased respiration rates can result in a higher energy deficit, reduce biomass and yield. The equilibrium between rate of photosynthesis and respiration are crucial for maintaining energy equilibrium in crops.

At a specific temperature (shown as the intersection of the respiration and photosynthesis lines in **Figure 7**), the rates of respiration and photosynthesis are equal (compensation point). However, as the temperature increases beyond this point, respiration rate increases while photosynthesis rate decreases. This suggests that organic molecules are being consumed at a higher rate, leading to a decrease in crop yield. In other words, the plant's energy consumption surpasses its energy production at higher temperatures, resulting in reduced crop productivity.

Many studies confirmed that **respiration rates increase under elevated night temperature**. Mohammed and Tarpley (2009) revealed that the high night-time temperature increased respiration rates and negatively affected the yield (by 95%) of rice. Loka and Oosterhuis (2010) observed that the high temperature regimes (27°C

and 30°C) increased respiration rates significantly by 49% and 56% compared to the control (24°C). ATP levels decreased proportionally with increasing temperature, while carbohydrate content remained unaffected. Long-term high night temperatures (30/28°C) for four weeks also increased respiration (39% and 21%) and reduced ATP levels (38% and 37%). Sucrose and hexose content consistently decreased with high night temperatures. **In summary, high temperatures increased respiration, leading to reduced ATP levels and carbohydrate content in the leaves.**

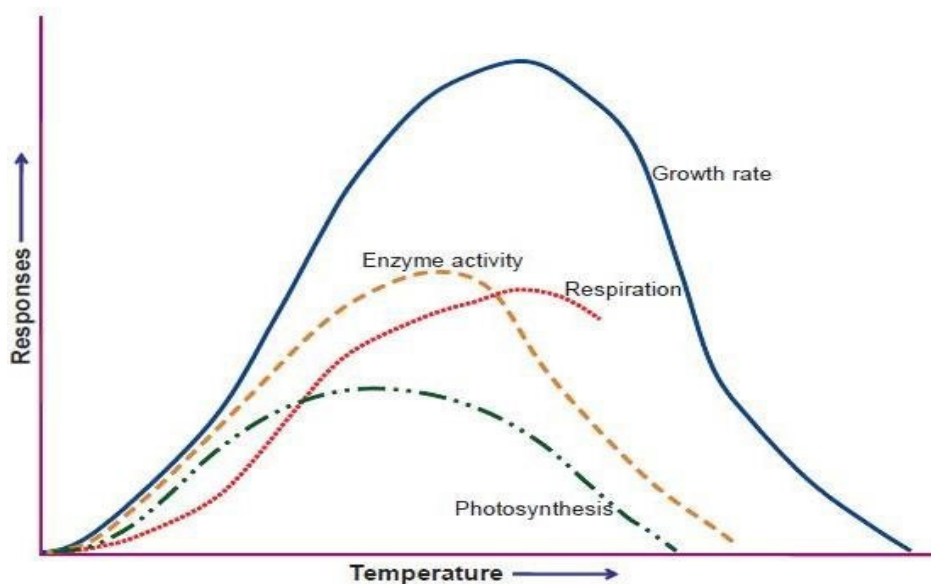


Figure 7: Schematic diagram of the effect of temperature on respiration, photosynthesis, enzymatic activity and growth rate of plants (Source: Fitter and hay, 2002).

4. Pollination and reproductive success

Temperature plays a vital role in pollination and reproductive success. Elevated temperatures have adverse effects on various stages of plant reproduction, including **flower initiation, pollen viability (germination and tube growth), stigma receptivity, ovule viability, ovule size, fertilization, seed/fruit set, seed composition, grain filling, and seed quality (Barnabás et al., 2008).**

High temperatures, particularly those exceeding 30°C, can significantly impact the development of male and female reproductive organs during the flowering stage (Lavania *et al.*, 2015). Heat stress negatively affects the reproductive stage of plants by causing a decrease in both the quantity and size of flowers, leading to

deformities in floral organs and subsequent loss of flowers and young pods. As a consequence, this results in a reduction in seed yield (Morrison and Stewart, 2002). Elevated temperatures can hinder fertilization by impeding the development of male and female gametophytes. This reduced fertilization is a common issue associated with heat stress, as it disrupts meiosis and fertilization in different plant species, ultimately leading to poor seed setting. During pollen development, exposure to high temperatures can hinder fertilization and seed development by reducing the quantity of mature pollen grains available for pollination. This is a result of abnormal pollen development, leading to a decrease in the viability (pollen viability is affected by the moisture content of the pollen, which is closely related to the vapor pressure deficit resulting from high temperatures.) and germinability of the remaining pollen grains, further impacting the plant's reproductive success (Jain *et al.*, 2007).

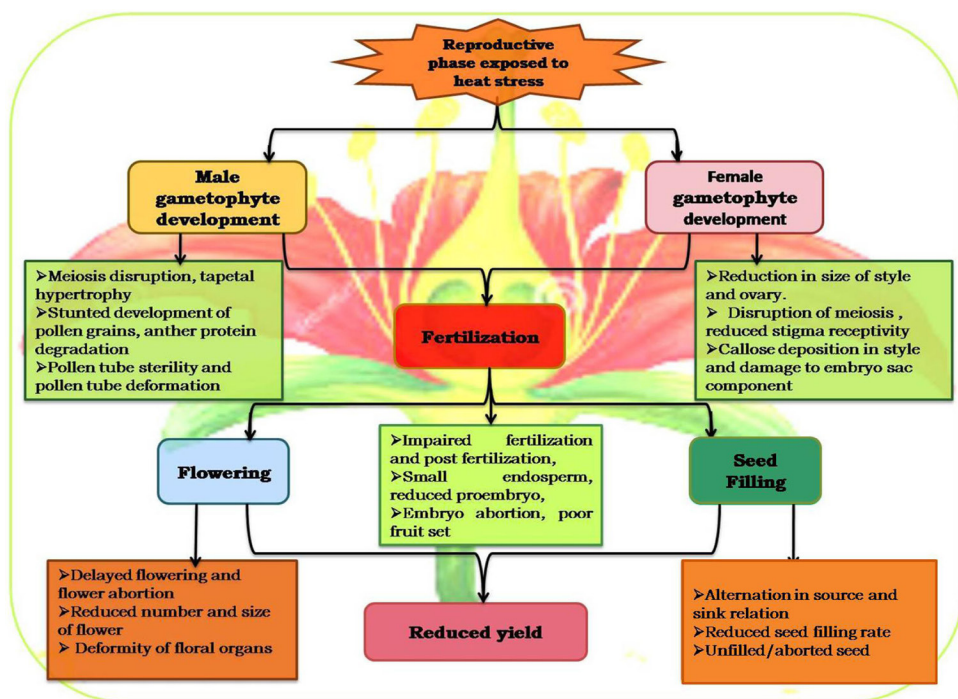


Figure 8: Effects of heat stress during the different reproductive phases (Source: Sita *et al.*, 2017).

5. Root physiology

Root size and morphology are vital for water and nutrient uptake. The root system requires an optimal temperature for growth, as excessive temperatures can disrupt uptake processes. The root system generally maintains a lower temperature compared

to the shoot. Despite occupying the same environment, different plant species may exhibit varying root system architectures (RSA) due to the influence of soil temperatures. Moreover, the optimal root temperature can vary among different plant species.

Luo *et al.*, (2020) revealed that the root system architecture (RSA), particularly root depth and root width, varied significantly among the species in response to higher temperatures. However, the aboveground shoot traits responded similarly across the species. The study also observed that increased temperature had little impact on the allocation of resources between the shoot and root. Under high temperatures, negative root interactions were intensified, and species with larger root size and fast early root expansion had a competitive advantage. In conclusion, the study suggests that plants with greater root resilience are better able to adapt to high temperatures. The differences in root interactions also indicate that changes in temperature could alter plant competition. Seedlings with larger root size and rapid early root expansion may have an advantage in adapting to warmer climates.

During root respiration, cells take up oxygen from the air present in the spaces among soil particles in the root zone. The uptake of oxygen increases with every 10°C rise in soil temperature, as oxygen solubility is inversely related to temperature. This higher oxygen demand can lead to root **hypoxia, a condition where roots experience insufficient oxygen supply** (Fonseca *et al.*, 2021). In general, high temperatures can have varied effects on the growth of plant roots. It can be promotive, inhibitory, or initially stimulatory and then become inhibitive once the optimum temperature is surpassed.

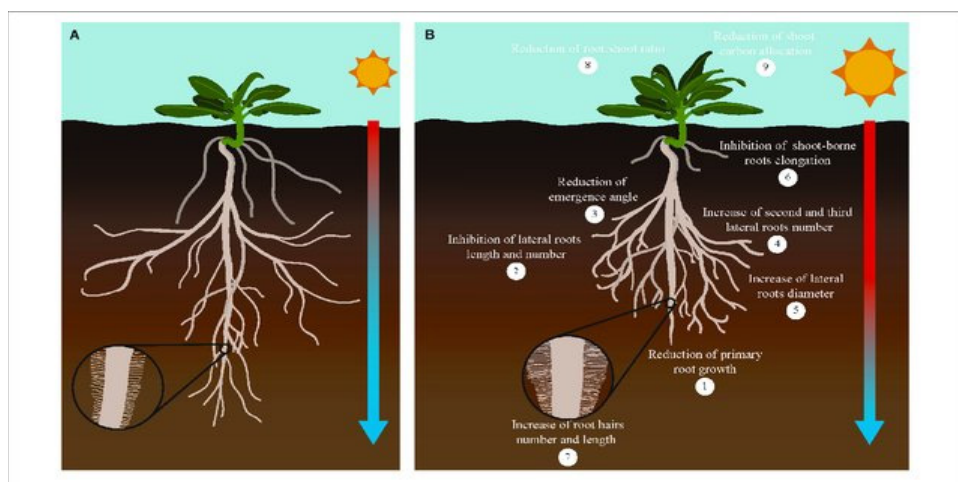


Figure 9: Response of major root traits to increasing temperatures in crops (Root system architecture is drastically altered in response to increased temperature

in the soil (B) compared to plants growing in optimal conditions (A). Crops growing under higher temperatures show shorter primary roots (1), reduction of lateral roots growth and number (2) and their angle of emergence (3), higher number of second and third order roots (4) with larger diameter (5), inhibition of shoot-borne roots (adventitious and nodal roots) elongation and number (6) and increase of root hairs number and length (7) (Source: Calleja-Cabrera et al., 2020).

6. Stomatal conductance

The water status in the leaf, stomatal conductance, and intercellular CO₂ concentration are influenced by high temperatures. Stomatal conductance plays a crucial role in regulating the diffusion of CO₂ from the atmosphere into the sub-stomatal cavity. It is a critical factor that significantly influences the rate of photosynthesis and the overall carbon metabolism of plants (Wang *et al.*, 2020). Studies show conflicting results on the effect of warming on stomatal conductance: some indicate increased conductance, while others suggest closure or no significant change. Understanding the combined effects of elevated CO₂ and temperature on stomatal conductance is crucial for plant carbon metabolism (Wang *et al.*, 2020). Many studies suggested that the stomatal conductance is decreased under elevated CO₂ condition. Increased stomatal conductance and related leaf cooling contribute to heat tolerance against high temperature.

Influence of CO₂ on Crop Performance

The influence of CO₂ on crop performance is significant and multifaceted. The Earth's atmosphere is experiencing a rise in carbon dioxide (CO₂) levels (<https://www.co2.earth/daily-co2>), leading to a potential increase in the greenhouse effect and future air temperatures. As a result, plants are responding to these changes in CO₂ concentration and temperature. Consequently, climate change poses a significant risk to agriculture as it may impact crop performance and overall agricultural productivity. Here some discussion made on effect of elevated CO₂ on plants response.

» Effect of elevated CO₂ on plant growth

Carbon dioxide serves as a vital link between the atmosphere and the biosphere and acts as a crucial substrate for the process of photosynthesis. When CO₂ levels are elevated, it stimulates photosynthesis, resulting in a higher uptake and assimilation of carbon (C) by plants. As a consequence, this increased carbon uptake leads to enhanced plant growth.

Differences in CO₂ utilization during photosynthesis lead to varying growth responses in plants. Those with a C3 photosynthetic pathway tend to show a more significant growth response compared to plants with a C4 pathway under CO₂

elevated condition (Prior *et al.*, 2011). C3 species generally exhibit a more positive response to increased CO₂ levels compared to C4 species. This is because C3 species experience a significant increase in their photosynthetic rate, approximately around 58%, when the CO₂ concentration is doubled (Drake *et al.*, 1997). On the other hand, C4 species' photosynthesis tends to reach a saturation point under current ambient CO₂ levels, meaning they may not show a substantial increase in response to further CO₂ elevation.

Conversely, under conditions of elevated temperature and drought, C4 species are predicted to have an advantage over C3 species (Long 1999). This is because C4 species possess a **CO₂ concentrating mechanism (CCM)** (special carbon fixation pathway that allows them to efficiently capture and concentrate CO₂ in specific cells within their leaves) that allow to these plants to concentrate and store CO₂ in their cells, allowing them to continue photosynthesis efficiently even when atmospheric CO₂ concentrations are low or when stomata are partially closed to reduce water loss (Figure 10). As a result, C4 species are better adapted to cope with water scarcity and higher temperatures, giving them a competitive edge over C3 species in such challenging environments.

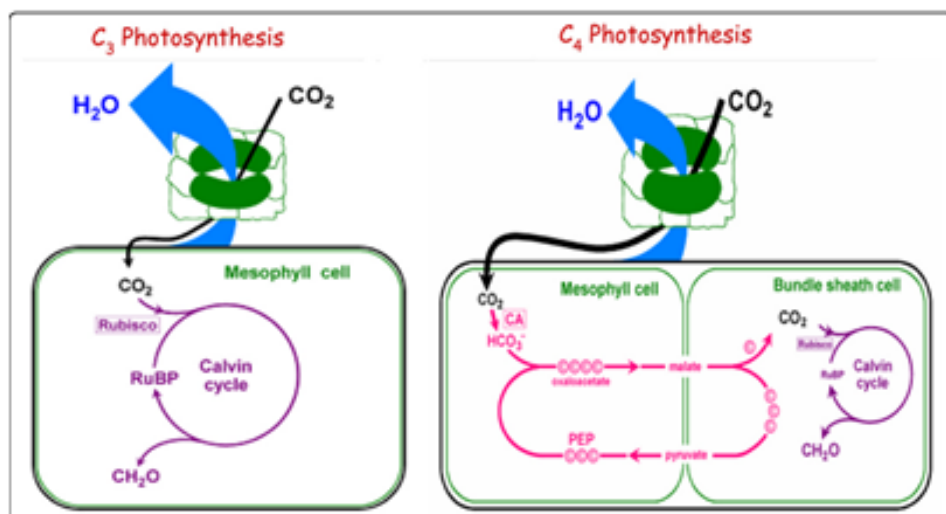


Figure 10: A schematic diagram of C3 and C4 photosynthesis (Source: Wang *et al.*, 2012).

Furthermore, under elevated CO₂ conditions, there is evidence of increased root growth and enhanced colonization of plant roots by mycorrhizal fungi. Mycorrhizae form a beneficial symbiotic association with plant roots, and this interaction is found to be more pronounced when CO₂ levels are elevated (Azmat *et al.*, 2015) (**Figure**

11). Mycorrhizae have been shown to enhance nutrient uptake by their host plants, as well as provide additional water to plants by proliferating hyphae in the soil which is useful under water stress condition.

Besides stimulating photosynthesis and aboveground growth, elevated CO_2 levels can also influence the allocation of carbon (C) within plants. The increased carbon supply resulting from elevated CO_2 concentrations can lead to a preferential distribution of photosynthate belowground (Prior *et al.*, 1997). In many instances, a significant portion of the extra biomass produced under elevated CO_2 is allocated belowground, resulting in an increased root-to-shoot ratio.

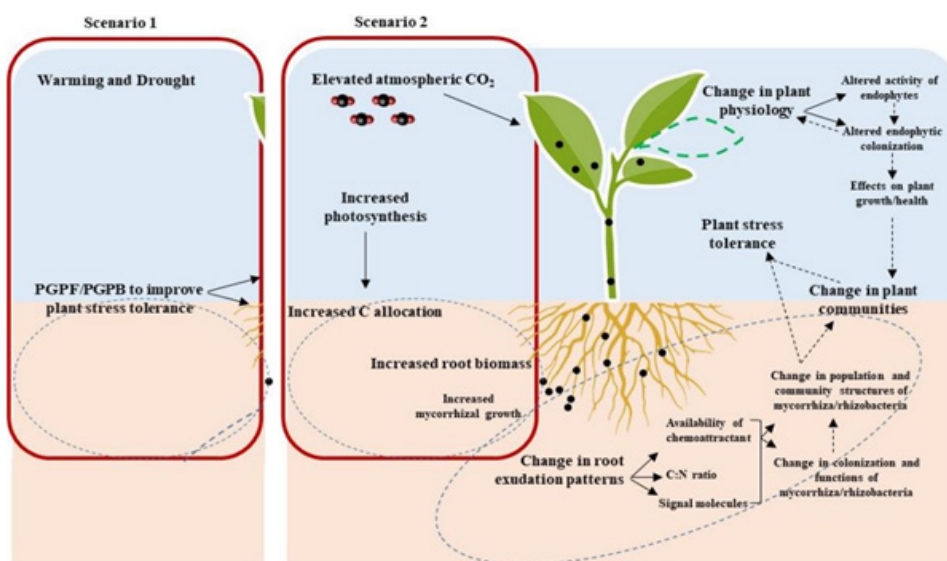


Figure 11: The effects of climate changes associated with elevated CO_2 , warming and drought events on the plant-microbial interactions in the rhizosphere (Source: Biswas *et al.*, 2019).

» Effect of elevated CO_2 on plant water relations

Elevated CO_2 levels have a negative impact on stomatal conductance (closure of stomata), which refers to the rate of material passage through stomata. This is mainly due to two effects: a reduction in the number of stomata per unit of leaf area and a decrease in stomatal apertures. As a consequence, **stomatal conductance decreases, resulting in reduced transpiration**, which is the loss of water from leaves (**improve water use efficiency**). This reduction in transpiration is advantageous for plant water conservation (Keenan *et al.*, 2013).

But on other hand, under elevated CO_2 levels, reduction of stomatal apertures

and density, leading to **reduced heat evapotranspirative cooling capacity of leaves**. This can exacerbate heat stress, particularly during water shortage, as the plants have limited ability to dissipate heat through transpiration, potentially causing damage to cellular structures and metabolism and ultimately impact on crop yield (Lond and Ort, 2010). **Figure 12** illustrates that under heat stress conditions alone, stomata remain open, leading to continued water loss. However, when heat stress is combined with elevated CO_2 conditions, stomata partially close due to the production of H_2O_2 , which helps reduce guard cell turgidity and consequently results in less water loss under heat stress conditions.

In this study, we focus solely on the impact of elevated CO_2 levels on crop performance. However, in real-world conditions, as CO_2 concentrations increase, there is a parallel rise in temperature because CO_2 is a greenhouse gas. Therefore, it becomes crucial to investigate the combined effect of both elevated CO_2 and temperature on crop performance. Considering the interactive effects of these two factors is essential to gain a comprehensive understanding of how climate change may influence crop growth, yield, and overall agricultural productivity.

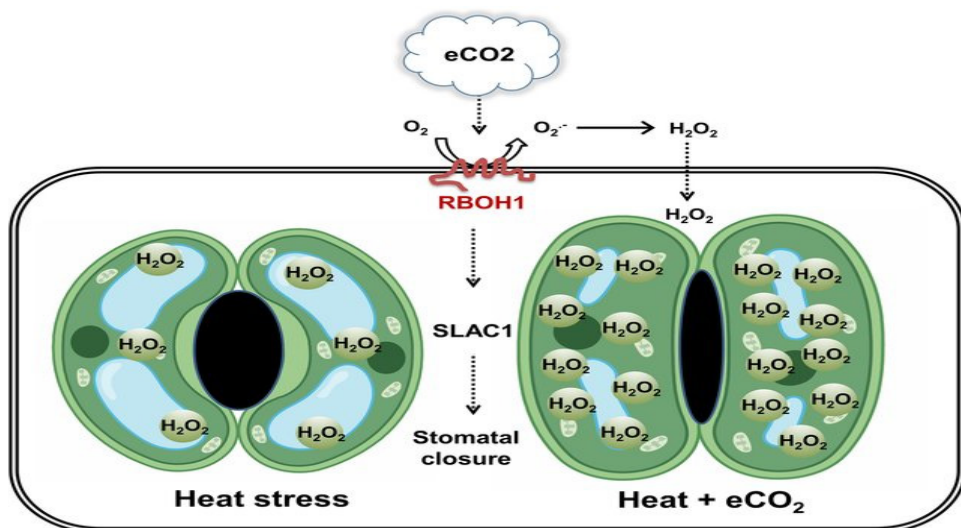


Figure 12: Elevated CO_2 induces thermotolerance by promoting stomatal closure through H_2O_2 production and SLAC1 upregulation, enhancing heat stress resistance (Source: Ahammed et al., 2021).

Conclusion

Environmental factors, including light, temperature, and CO₂ levels, have significant and interconnected effects on crop performance. Light serves as the primary source of energy for photosynthesis, regulating various physiological processes and photomorphogenic responses in plants. Temperature plays a critical role in germination, growth, and reproductive success, with both positive and negative impacts depending on the crop and the range of temperatures experienced. Elevated CO₂ levels stimulate photosynthesis and plant growth, particularly in C3 species, while affecting root growth and mycorrhizal interactions. However, elevated CO₂ can also lead to stomatal closure, reducing transpiration and potentially exacerbating heat stress under water shortage conditions.

The interactive effects of these environmental factors are complex and must be considered to understand their combined impact on crop physiology. As the planet faces rising CO₂ levels and temperatures due to climate change, it is essential to study the combined effect of elevated CO₂ and temperature on crop performance to develop sustainable agricultural practices. Farmers and researchers should aim to optimize light exposure, temperature control, and CO₂ levels to enhance crop productivity while minimizing the negative impacts of environmental stressors. Understanding the specific requirements of different crops and managing environmental factors accordingly will be crucial for ensuring food security and sustainable agriculture in a changing climate.

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